AERO5560 Flight Mechanics 2

Group Project: Aircraft Closed-Loop Control System Design and Analysis

Introduction

This assignment builds on the analyses performed in the individual assignment. The objective is to design a complete autopilot, investigate the stability and performance of the control system, and study the change in response of the aircraft to gust inputs.

Submission Instruction

You must submit a final report and your source code to the corresponding Canvas portal by the due date to avoid late penalties. The longitudinal and lateral-directional subgroups must combine your report into a single PDF file that is coherent and well-formatted. **The page limit of the final report is 60 pages.** A report template is provided on Canvas. You do not need to follow the template exactly. The template provides useful guidelines on how to format your report and what contents you should put in the report to attract a higher grade. The source code should be submitted as a single ZIP file, and it must be stand-alone (meaning we can run your code without importing additional files).

General Instructions

You are to work in groups of 6-8, with each group considering a different aircraft. Within each group, 3-4 members will consider the lateral-directional problem, and the other 3-4 will consider the longitudinal problem. The lateral subgroup will aim to design the heading-hold guidance loop and yaw damper, whereas the longitudinal subgroup's goal is to design the vertical speed guidance loop and auto-throttle control.

You are to decide amongst yourselves who will work on what problem. The two sub-groups will be mostly independent from one another when designing the individual control loops, but are expected to work closely together on the final controller integration.

There are three parts of the group project:

- Longitudinal Subsystem Tasks worth 80 points (should be completed by the longitudinal subgroup, and the marks allocated to these tasks will only be awarded to the longitudinal subgroup members).
- Lateral-Directional Subsystem Tasks worth 80 points (should be completed by the lateral subgroup, and the marks allocated to these tasks will only be awarded to the lateral subgroup members).

- Complete Autopilot Integration worth 20 points (should be completed by both subgroups together, and the marks will be awarded equally to the two subgroups).

The design is to be done using Bode and Root Locus design methodologies. Ensure that you illustrate your control loop topologies, and that you write down and explain how you have formulated your closed-loop transfer functions. Ensure that all transfer functions presented are given in Zero-Pole-Gain (ZPK) form. You should also present the transfer functions in terms of sensitivity functions, complementary sensitivity functions, and compensator transfer functions where appropriate (SCK form). You may reuse any functions you have developed as part of your MATLAB Task Group 1 and 2 submissions.

Peer Assessment

Self and peer assessment are required for this group-based assignment. We will use Spark Plus, detailed instructions will be provided in the lectures.

Longitudinal Subsystem Tasks (80 points)

The objective is to develop a vertical speed autopilot with auto-throttle that will make the aircraft track a specified vertical speed by using the elevator actuator while regulating airspeed with throttle.

1. **[15] Inner Loop: Pitch Rate Autopilot**

   To overcome the inherent problem in using the elevator to directly control vertical speed, you need a pitch rate autopilot that seeks to use the elevator to control pitch rate. The final vertical speed guidance loop can then use pitch angle to control vertical speed.

   - Determine $G^q_{\delta_e}$ of the supplied aircraft.
   - Construct a feedback loop that controls pitch rate q using the elevator and illustrate the block diagram. Label all the blocks and signals.
   - Use MATLAB sisotool to design a compensator that can achieve the following specifications (for the time domain specifications use a $4°/s$ pitch rate step input):
     - Crossover frequency ω_c (-3 dB point) in the region of 2 to 10 rad/s (you may vary this range depending on your findings and depending on the airspeed).
     - Noise bandwidth of 20 rad/s
     - Settling time: 3 s
     - Steady state error: less than 1 %
     - Maximum overshoot: less than 20 %
     - Elevator deflection does not exceed the limit.
   - You must investigate at least **THREE** options for the compensator. Document the evolution of your design. For each design iteration, show the compensator transfer function and how the responses improve with each design iteration using root locus, Bode plot, and step response. Tabulate the key responses and compare them to the specifications.
   - Explain your design approach and justify your choices. Discuss whether there is any room for improvement for your final design.

2. **[15] Outer Loop: Vertical Speed Guidance Loop**

   Design a vertical speed guidance loop that incorporates the pitch rate autopilot as an inner loop. The guidance loop should use a compensator in the forward leg of the loop to convert vertical speed error into a pitch rate command (q_c).

   - Construct the vertical speed guidance loop incorporating the pitch rate autopilot and illustrate the block diagram. Label all the blocks and signals.
   - Identify the plant of the vertical speed guidance loop and show the plant transfer function.
   - Use MATLAB sisotool to design a compensator that can achieve the following specifications (for the time domain specifications, use a 500 ft/min vertical speed step input):

  - Crossover frequency ω_c (-3 dB point) in the region of 0.5 to 2 rad/s (you may vary this range depending on your findings and depending on the airspeed).
  - Noise bandwidth of 20 rad/s
  - Rise time: 3 s
  - Settling time: 10 s
  - Steady state error: less than 1 %
  - Maximum overshoot: less than 20 %
  - Elevator deflection does not exceed the limit.
- Discuss your design approach. You don't need to show the design iteration for the vertical speed loop compensator.
- Show the frequency response of the vertical speed guidance closed-loop system. Discuss whether the response has met all the specifications, and identify any potential problems.
- Show the step response of the vertical speed to a 500 ft/min vertical speed step input. Discuss whether the response has met all the specifications, and identify any potential problems.
- Show the step response of the inner loop pitch rate to a 500 ft/min vertical speed step input. Discuss whether the response is reasonable, and identify any potential problems.
- Show the step response of the inner loop elevator deflection to a 500 ft/min vertical speed step input. Discuss whether the response is reasonable, and identify any potential problems.

3. **[15] Vertical Speed Guidance Loop Gust Response**

   As you have the vertical speed guidance loop design now, use the gust models and PSD methods introduced in this course to analyse the system response to gust input. You do not need to present the PSDs and PDFs. Tabulate the results for easier comparison.

   - Develop the closed-loop transfer functions relating the responses of pitch rate, vertical speed, and elevator to disturbances. You must consider both vertical speed and pitch rate disturbances (a total of 6 transfer functions). Express the transfer functions in both SCK and ZPK forms.
   - Use the Dryden gust models to develop the relevant PSDs and PDFs of the open-loop aircraft pitch rate and vertical speed response to gusts. Determine the maximum likely open-loop pitch rate and vertical speed excursion due to each longitudinal gust component, and then the corresponding maximum likely total excursions.
   - Use the same gust models, develop the relevant PSDs and PDFs of the closed-loop aircraft pitch rate and vertical speed response to gusts. Determine the magnitudes of the maximum likely excursion due to each longitudinal gust component and compare them to the open-loop maximum excursions. Comment on the effectiveness of the guidance loop design at rejecting gusts. You should tabulate the results.

4. **[20] Auto-Throttle Design**

   Design an auto-throttle control system. This loop will convert airspeed error into a throttle command. The design procedure is similar to Task 2.

   - Construct the complete MIMO loop topology of the longitudinal autopilot system, incorporating the pitch rate autopilot, the vertical speed guidance loop, and the auto-throttle. Note that the vertical speed loop and the auto-throttle loop are cross-coupled. The effect of the throttle on pitch rate is small and can be ignored.
   - Develop the new cross-coupled plant transfer functions $G^{u}_{\delta_T MIMO}$, and $G^{V_s}_{\delta_e MIMO}$ for the complete MIMO system. Present the transfer functions in SCK form.
   - Design the compensator to the same specifications as Task 2. Ensure that you evaluate the plant dynamics with the vertical speed system coupling effects considered. The compensator should be designed for this plant. Show the final compensator transfer function.
   - Check if the pitch rate autopilot and the vertical speed guidance loop compensator should be redesigned. If necessary, redesign them to the same specifications in tasks 2 and 3.

5. **[15] Longitudinal MIMO System Response**

   Analyse the longitudinal MIMO system response to primary control inputs and gust disturbances. The complete system includes the pitch rate controller, the vertical speed guidance loop, and the auto-throttle.

   - Develop the transfer functions ($G^{u}_{u_c MIMO}$ and $G^{v_s}_{v_{s_c} MIMO}$) that describe the primary control effects of the MIMO system. Quantify the primary control effects in time domain (i.e. rise time, overshoot, steady state error, settling time of a step response), and compare them to the responses obtained without the secondary cross-coupling effects ($G^{u}_{u_c SISO}$ and $G^{v_s}_{v_{s_c} SISO}$). Comment on the overall effectiveness of the MIMO system at tracking a reference compared to the SISO system.
   - Develop the transfer functions ($G^{v_s}_{u_c MIMO}$ and $G^{u}_{v_{s_c} MIMO}$) that describe the secondary effects of the MIMO system. Quantify the secondary effects in time domain (i.e. rise time, overshoot, steady state error, settling time of a step response), and compare them to the responses obtained without the cross-coupling effects ($G^{v_s}_{u_c SISO}$ and $G^{u_c}_{v_{s_c} SISO}$). Comment on the overall effectiveness of the MIMO system at minimising the secondary effects compared to the SISO system.

Lateral-Directional Subsystem Tasks (80 points)

The objective is to develop a heading-hold autopilot that will make the aircraft seek and track a specified heading (Yaw angle) by using the aileron. A yaw damper must also be designed to improve the Dutch Roll dynamics.

1. **[15] Inner Loop: Wing-Leveller**

   To overcome the inherent problem in using aileron to directly control heading, you need a wing-leveller which seeks to use aileron to hold bank angle. The final heading hold guidance loop can then use bank angle to control heading.

   - Determine $G_{\delta_a}^{\phi}$ of the supplied aircraft.
   - Construct a feedback loop that controls bank angle using aileron and illustrate the block diagram. Label all the blocks and signals.
   - Use MATLAB sisotool to design a compensator that can achieve the following specifications (for the time domain specifications use a 30° bank angle step input):
     - Crossover frequency ω_c (-3 dB point) in the region of 2 to 10 rad/s (you may vary this range depending on your findings and depending on the airspeed).
     - Noise bandwidth of 20 rad/s
     - Settling time: 3 s
     - Steady state error: less than 1 %
     - Maximum overshoot: less than 20 %
     - Aileron deflection does not exceed ±20°
   - You must investigate at least **THREE** options for the compensator. Document the evolution of your design.
   - For each design iteration, show the compensator transfer function and how the responses improve with each design iteration using root locus, Bode plot, and step response. Tabulate the key responses and compare them to the specifications.
   - Explain your design approach and justify your choices. Discuss whether there is any room for improvement for your final design.

2. **[15] Outer Loop: Heading Hold Guidance Loop**

   Design a heading hold guidance loop that incorporates the wing-levelling autopilot as an inner loop. The guidance loop should use a compensator in the forward leg of the loop to convert heading (Yaw angle) error ($\Delta\Psi$) into a bank angle command (ϕ_c).

   - Construct the heading hold guidance loop incorporating the wing-leveller and illustrate the block diagram. Label all the blocks and signals.
   - Identify the plant of the heading hold guidance loop and show the plant transfer function.
   - Use MATLAB sisotool to design a compensator that can achieve the following specifications (for the time domain specifications use a 30° heading angle step input):

  - Crossover frequency ω_c (-3 dB point) in the region of 2 to 10 rad/s (you may vary this range depending on your findings and depending on the airspeed).
  - Noise bandwidth of 20 rad/s
  - Settling time: 10 s
  - Steady state error: less than 1 %
  - Maximum overshoot: less than 20 %
  - Aileron deflection does not exceed $\pm 20^\circ$
- Discuss your design approach. You don't need to show the design iteration for the heading hold loop compensator.
- Show the frequency response of the heading hold guidance closed-loop system. Discuss whether the response has met all the specifications, and identify any potential problems.
- Show the step response of the heading hold guidance closed-loop system to a 30° heading angle step input. Discuss whether the response has met all the specifications, and identify any potential problems.
- Show the step response of the inner wing-leveller output bank angle to a 30° heading angle step input. Discuss whether the response has met all the specifications, and identify any potential problems.
- Show the step response of the aileron to a 30° heading angle step input. Discuss whether the response has met all the specifications, and identify any potential problems.

3. **[15] Heading Hold Guidance Loop Gust Response**

   As you have the heading hold guidance loop design now, use the gust models and PSD methods introduced in this course to analyse the system response to gust input. You do not need to present the PSDs and PDFs. Tabulate the results for easier comparison.

   - Develop the closed-loop transfer functions relating the responses of bank angle, heading angle, and aileron to disturbances. You must consider both bank angle and yaw angle disturbances (a total of 6 transfer functions). Express the transfer functions in both SCK and ZPK forms.
   - Use the Dryden gust models to develop the relevant PSDs and PDFs of the open-loop aircraft bank angle and heading angle response to gusts. Determine the maximum likely open-loop bank angle and yaw angle excursion due to each lateral gust component, and then the corresponding maximum likely total excursions.
   - Use the same gust models, develop the relevant PSDs and PDFs of the closed-loop aircraft bank angle and heading angle response to gusts. Determine the magnitudes of the maximum likely excursion due to each lateral gust component and compare to the open-loop maximum excursions. Comment on the effectiveness of the guidance loop design at rejecting gusts. You should tabulate the results.

4. **[20] Yaw Damper Design and Redesigning Compensators**

   Design a classical yaw damper $\delta_r = K\frac{s}{s+a}r$ to improve the Dutch Roll dynamics. With the yaw damper added to the control system, the compensators you have designed in Tasks 2 and 3 may behave differently and may need to be redesigned. You will evaluate the new plant transfer functions and redesign the compensators.

- Construct the complete MIMO loop topology of the lateral autopilot system, incorporating the wing leveller, the heading hold guidance loop, and the yaw damper. Note that the heading hold guidance loop and the yaw damper loop are cross-coupled.
- Develop the new cross-coupled plant transfer functions $G^{r}_{\delta_r MIMO}$, $G^{\phi}_{\delta_a MIMO}$, and $G^{\psi}_{\delta_a MIMO}$ for the complete MIMO system. Present the transfer functions in SCK form.
- Design the yaw damper compensator. Show the final compensator transfer function.
- Evaluate the new plant transfer functions. Comment on the stability of the complete MIMO system and discuss whether a new design is necessary.
- Re-design the wing leveller compensator to the same specifications in task 2 using the new plant transfer function. Explain your design approach. (You don't need to document the iterations.) Show the final compensator transfer function.
- Re-design the heading hold guidance loop compensator to the same specifications in task 3 using the new plant transfer function. Explain your design approach. (You don't need to document the iterations.) Show the final compensator transfer function.

5. **[15] Lateral MIMO System Response**

   Analyse the lateral MIMO system response to primary control inputs and gust disturbances. The complete system includes the wing leveller, the heading hold guidance loop, the yaw damper, and the redesigned compensators.

   - Develop the transfer functions ($G^{\phi}_{\phi_c MIMO}$ and $G^{\psi}_{\psi_c MIMO}$) that describe the primary control effects of the MIMO system. Quantify the primary control effects in time domain (i.e. rise time, overshoot, steady state eroor, settling time of a step response), and compare them to the design responses obtained without the secondary cross-coupling effects ($G^{\phi}_{\phi_c SISO}$ and $G^{\psi}_{\psi_c SISO}$ using the compensator from Task 1 and 2). Comment on the overall effectiveness of the MIMO system at tracking a reference compared to the SISO system.
   - Develop the transfer functions ($G^{r}_{\phi_c MIMO}$ and $G^{r}_{\psi_c MIMO}$) that indicate the yaw damper's effect at rejecting unwanted yaw rate oscillation. Describe the yaw damper's performance in time domain, and compare them to the yaw rate responses obtained without the yaw damper activated ($G^{r}_{\phi_c SISO}$ and $G^{r}_{\psi_c SISO}$ using the compensator from Task 1 and 2). Comment on the overall effectiveness of the MIMO system at improving the Dutch Roll dynamics.

Complete Autopilot Integration (20 points)

The final step of this project is to implement both longitudinal and lateral controllers in the nonlinear simulation and assess their performance.

1. **[10] Complete Autopilot Performance** To verify that the controllers you have developed using the linear assumptions are realisable in the time domain, implement the control and guidance laws in the nonlinear simulation by representing your controllers as a state-space system.

   - Implement the complete autopilot in the non-linear simulation, including pitch rate autopilot, vertical speed guidance loop, auto-throttle, wing leveller, heading hold guidance loop and yaw damper. Explain how you implement the controllers in the time domain and the non-linear simulation code.
   - Command a vertical speed step input of 500 ft/min and plot the step responses of vertical speed, pitch rate, elevator deflection, airspeed, and throttle setting. Compare the responses to those obtained from the linear simulations. Discuss the key differences and what caused them.
   - Command a 30-degree heading angle step input and plot the step responses of heading, bank angle, aileron deflection, yaw rate, and rudder deflection. Compare the responses to those obtained from the linear simulations. Discuss the key differences and what caused them.
   - Discuss what problems can be expected if a large input is commanded (e.g. a large vertical speed input)? How could this be dealt with when implementing the control system? Demonstrate your approach in the non-linear simulation.

2. **[10] Performance in Off-Design Conditions**

   Trim the aircraft at flight conditions different to the initial conditions you have used so far, but maintain the same controller you have designed, and discuss how the performance of your controller changes at these different conditions. You can choose a couple of conditions (e.g. one condition at a faster airspeed, and one condition at a lower airspeed) and show the equivalent linear system responses. Study how the closed-loop poles move in the S-plane as you change the trim conditions.